

Load Balancers

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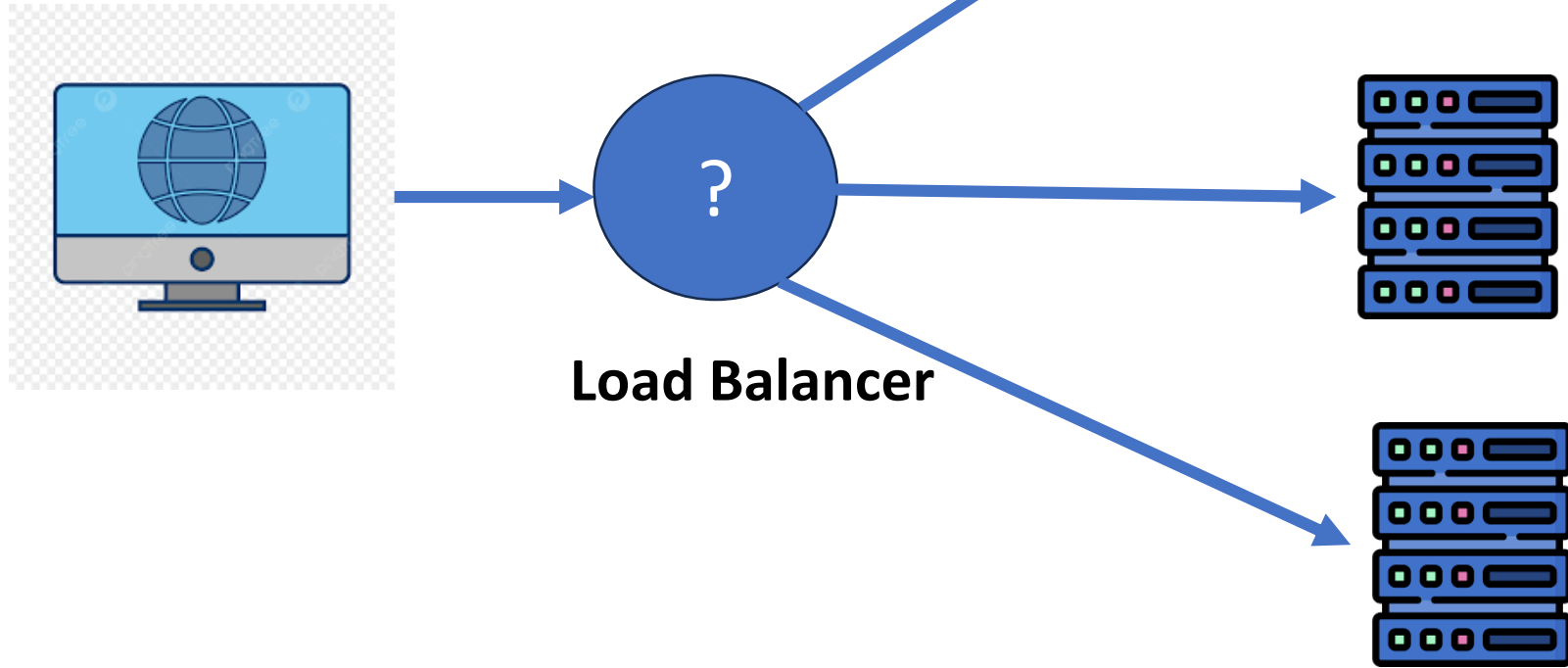
Load Balancer

- Single server setup



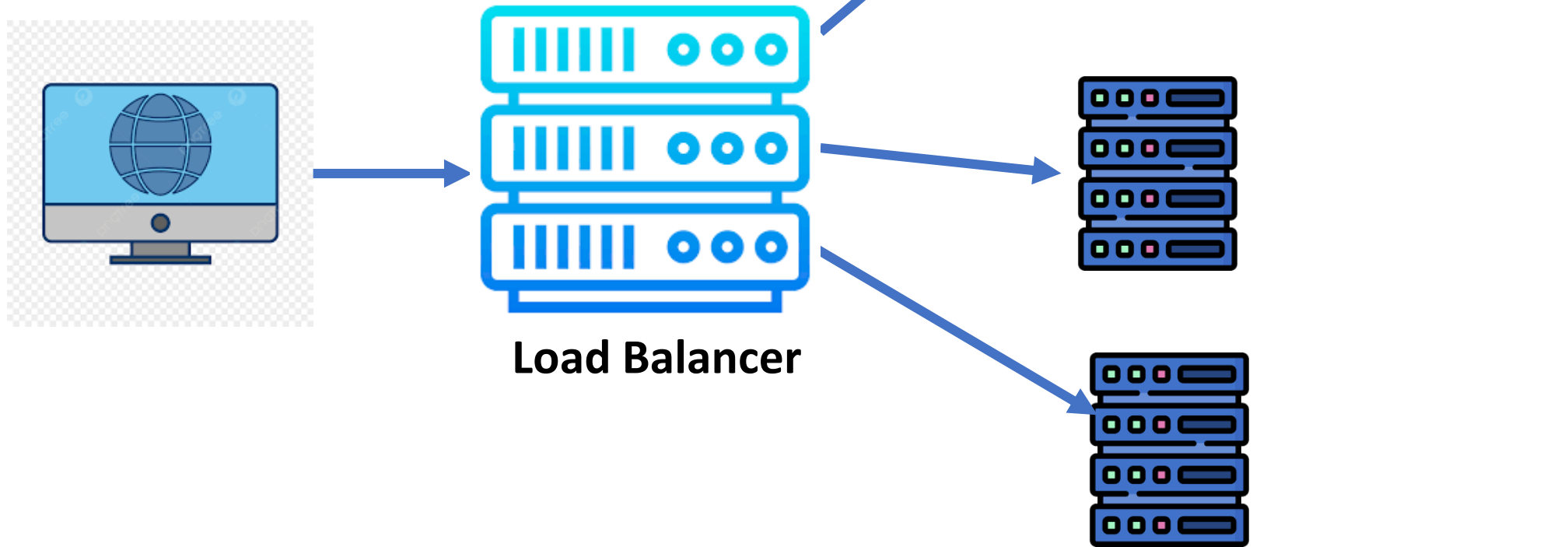
Load Balancer

- Multiple server setup



Load Balancer

- Multiple server setup



Load Balancer

Contd...

- Distribute incoming network traffic across multiple servers to ensure **no single server bears too much load.**
- Ensuring high availability, reliability, and faster application performance.
- Acting as a "traffic cop", it routes user requests to healthy servers using load balancing algorithms.

Health Check- Load Balancer

- Load balancer health checks are automated, periodic probes (HTTP/HTTPS/TCP) sent to **backend servers to verify** their availability and performance.
- Unhealthy instances are automatically removed from traffic rotation (marked "OutOfService" or "Unhealthy") and restored only when they pass subsequent checks.

Software Load Balancers

- NGINX (“EngineX”)

nginx.org → Docs → Admin’s Guide → Load Balancing

- HAPROXY

haproxy.com

Hardware Load Balancers

- F5

<https://www.f5.com/products/load-balancing>

- CITRIX

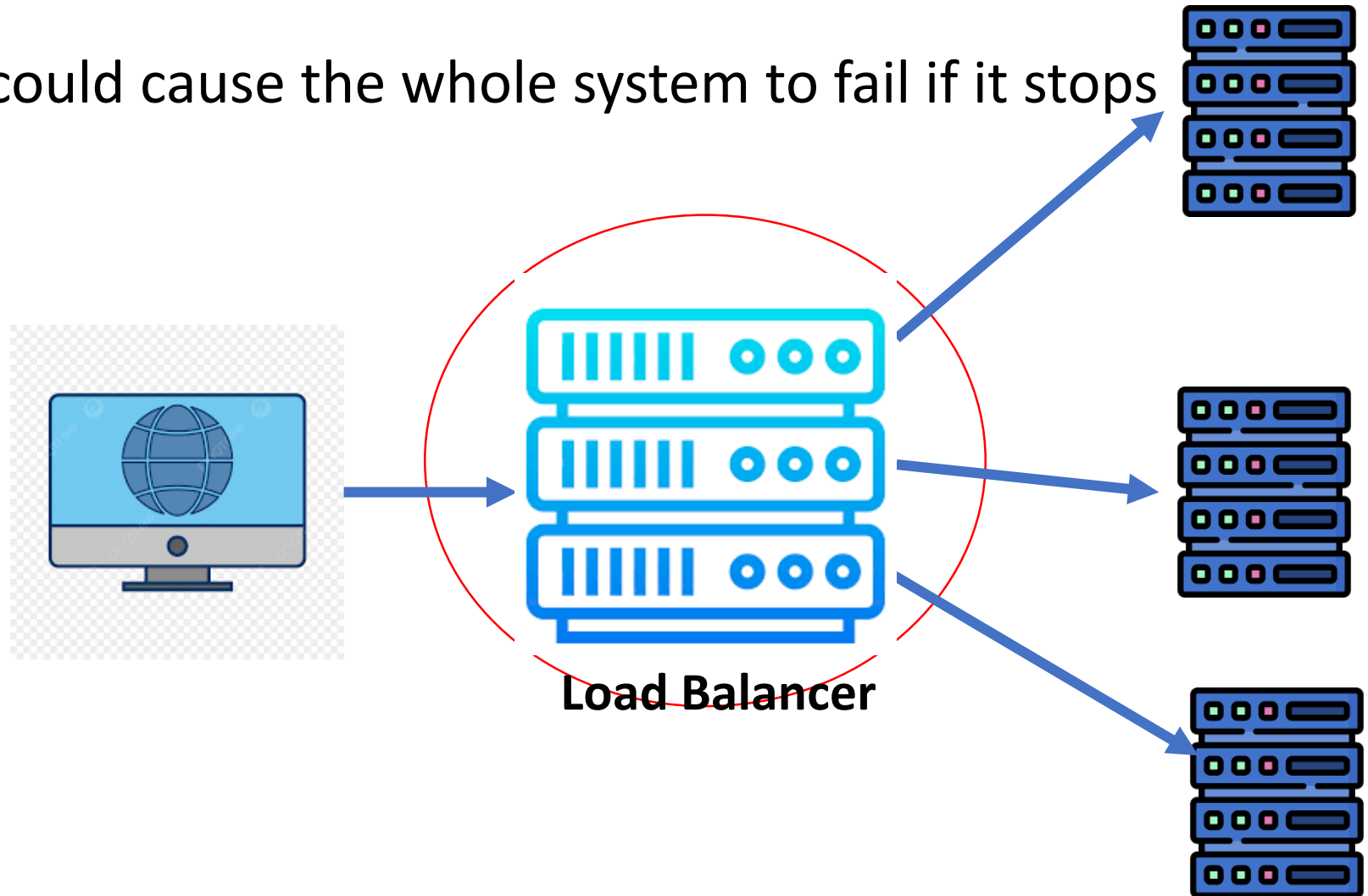
<https://docs.citrix.com/en-us/citrix-virtual-apps-desktops/install-configure/load-balance-machines.html>

Cloud Based Load Balancers

- **Google Cloud Load Balancing:** Offers global, **software-defined**, and fully distributed services.
- **AWS Elastic Load Balancing (ELB):** Includes **Application, Network, and Gateway** Load Balancers.
 - <https://aws.amazon.com/elasticloadbalancing/>
- **Azure Load Balancer:** Provides **high-performance, low-latency Layer 4 (Network Layer)** load balancing.

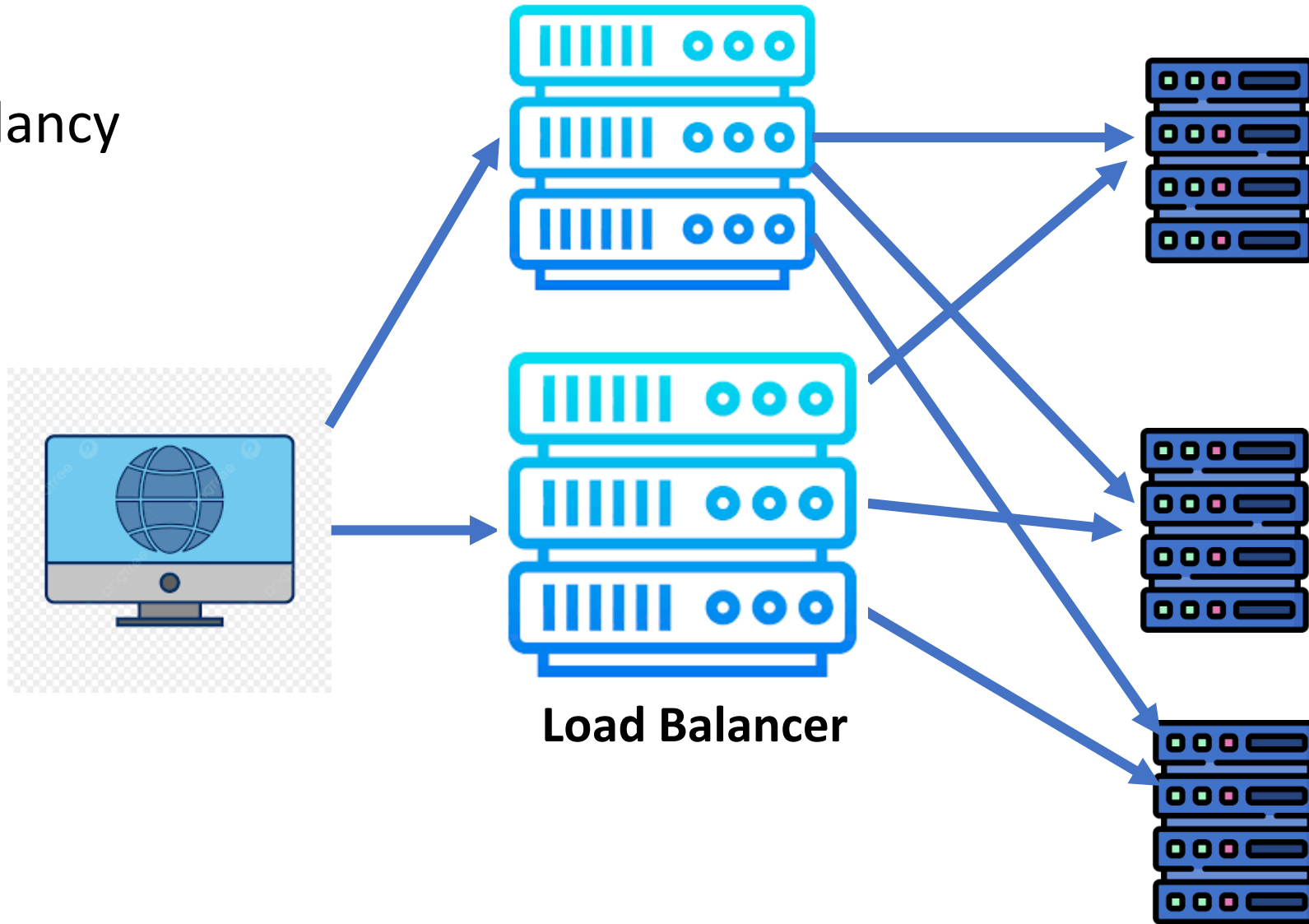
Single Point of Failure (SPOF)

- Any component that could cause the whole system to fail if it stops the working



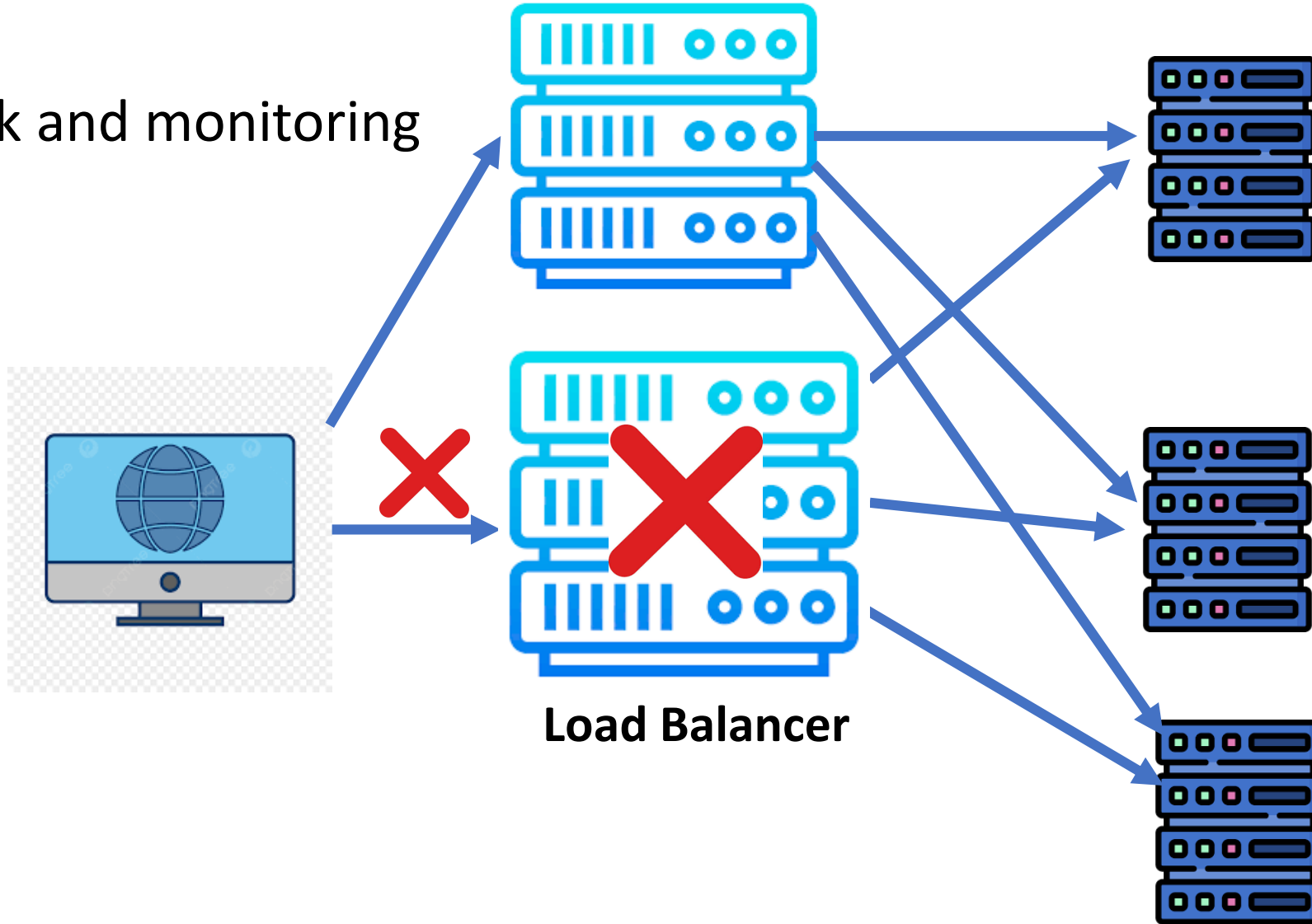
Strategies to Overcome SPOF

- Redundancy



Strategies to Overcome SPOF

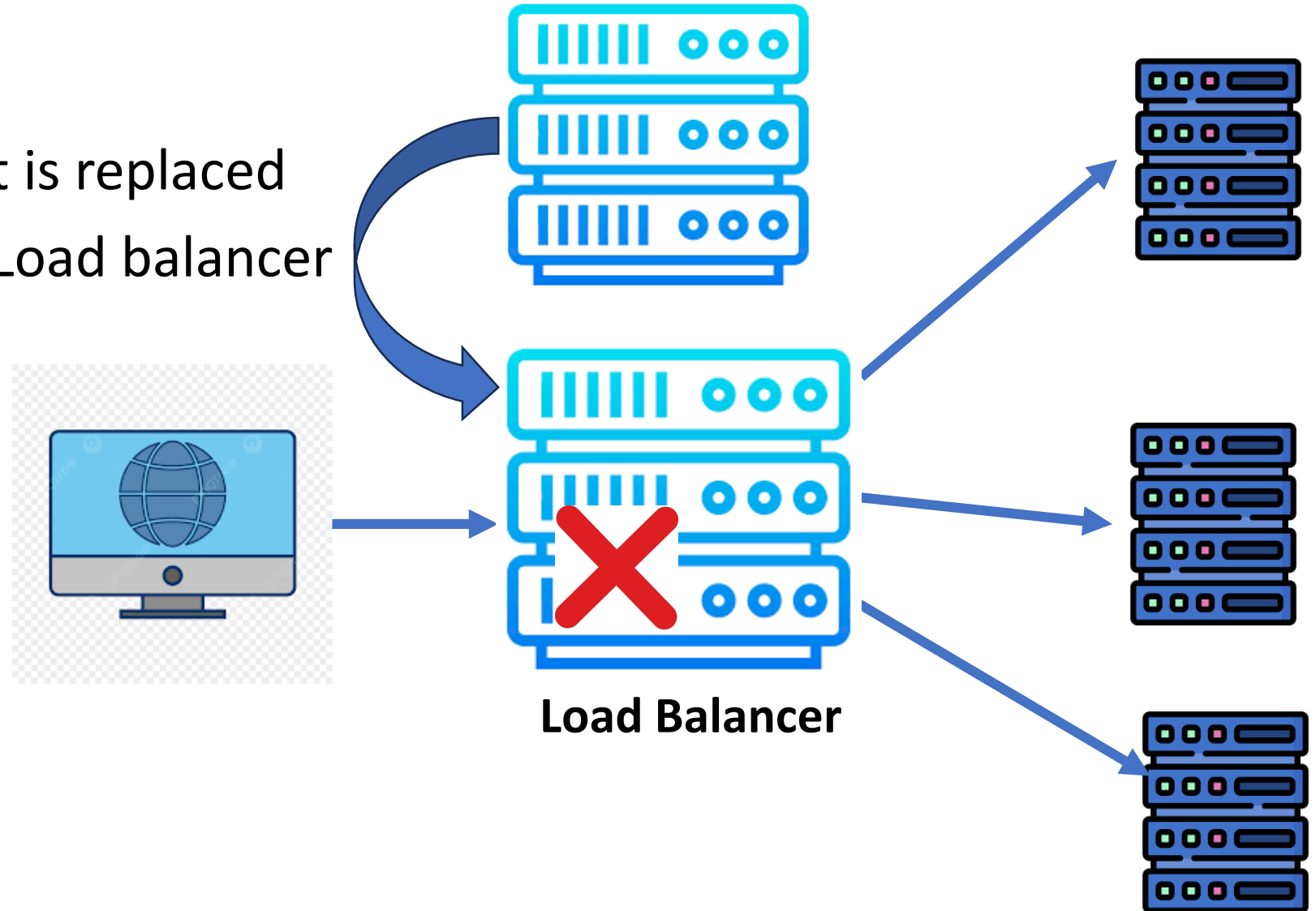
- Health check and monitoring



Strategies to Overcome SPoF

- Self Healing system

When load balancer fails it is replaced by new instance of same Load balancer



- Thank You